

Clustering Using Weka

Introduction:

Clustering involves identification of natural groups across a dataset. It is an unsupervised learning problem that deals with identifying structure in a collection of unlabeled data.

Background reading: Basic clustering concepts, like k-means

Dataset used: iris dataset

Clustering:

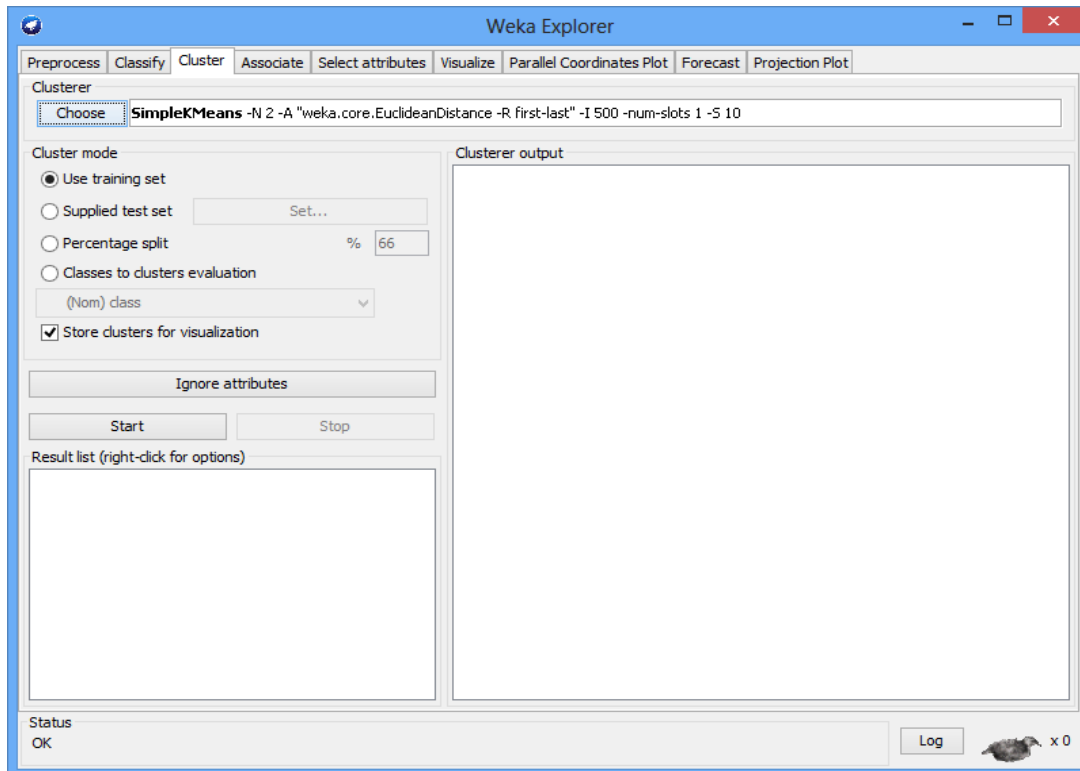
k-means:

The *k*-means algorithm is a simple, straightforward algorithm to assign instances to clusters. Each cluster is defined by a cluster centroid, and instances belong to the cluster for which their Euclidian distance to the centroid is the smallest. For each cluster a new centroid is found by taking the average over the cluster instances, which may lead to shifts of instances between clusters. This iterative process ends when the centroids stop changing.

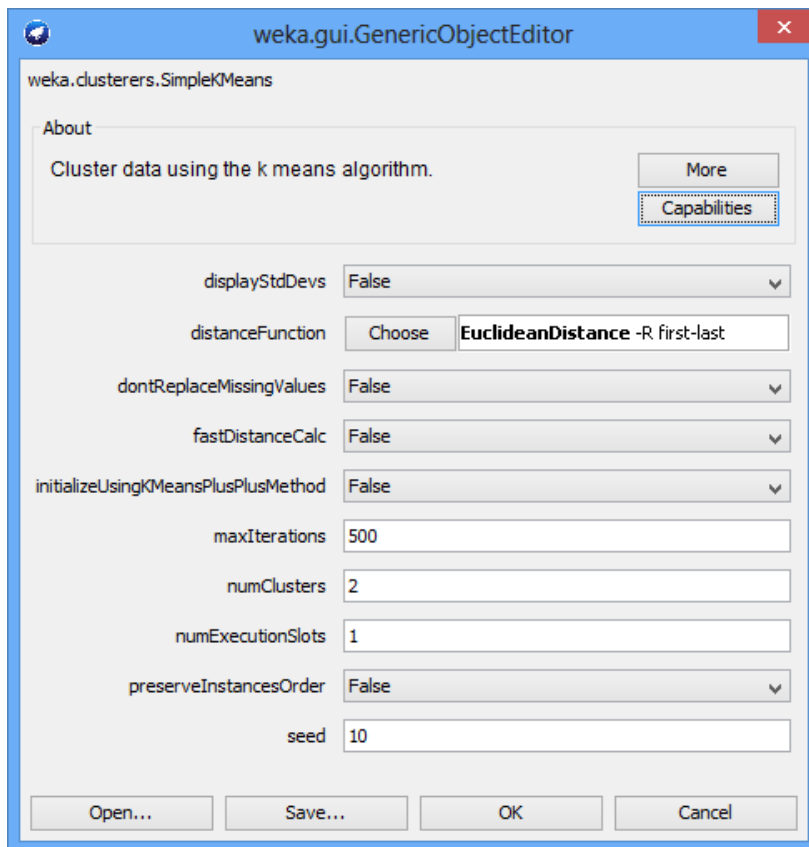
Start by preprocessing iris dataset in weka.

Navigate to cluster tab.

Click choose and select SimpleKmeans algorithm for clustering.



Now click anywhere in the textbox next to the choose button.



Click more to get details about the each parameter. The important ones to look for are distanceFunction, maxIterations and numClusters (value of k)

Specify the number of clusters as 3.

Output representation:

=== Run information ===

Scheme: weka.clusterers.SimpleKMeans -N 3 -A "weka.core.EuclideanDistance -R first-last" -I 500 - num-slots 1 -S 10

Relation: iris

Instances: 150

Attributes: 5

sepalength
sepalwidth
petallength
petalwidth
class

Test mode: evaluate on training data

=== Clustering model (full training set) ===

kMeans

=====

Number of iterations: 3

Within cluster sum of squared errors: 7.817456892309574

Missing values globally replaced with mean/mode

Cluster centroids:

Attribute	Cluster#			
	Full Data (150)	0 (50)	1 (50)	2 (50)
sepal.length	5.8433	5.936	5.006	6.588
sepal.width	3.054	2.77	3.418	2.974
petal.length	3.7587	4.26	1.464	5.552
petal.width	1.1987	1.326	0.244	2.026
class	Iris-setosa	Iris-versicolor	Iris-setosa	Iris-virginica

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0 50 (33%)

1 50 (33%)

2 50 (33%)

Visualization:

To look at the graphical representation. Right click on the model and select "Visualize cluster assignments"

